



SAFETY

- + Long life type prismatic LiFePO4 cells, suitable for energy storage application.
- + Low voltage system, safety for application.
- + Battery certification: UN38.3, MSDS, CE
- + Cycle life >4000 cycles.

DESIGN

- + Standard 19" rack design, flexible and easily installation.
- + -20~+55°C widely temperature range.
- + Parallel support for more energy or more capacity inverter.
- + SOC, RUN, ALARM display and BMS PC software.
- + OVP, LVP, OCP, OTP, LTP, Short circuit protection.
- + RS485, CAN communication port.
- + Independent protection for charge and discharge.
- + High reliability: AFE microchip is Panasonic & MCU is ST
- + Optional part: Bluetooth module, Heater.

Compatible with inverter brands

Communication compatible with major inverter



48V Series

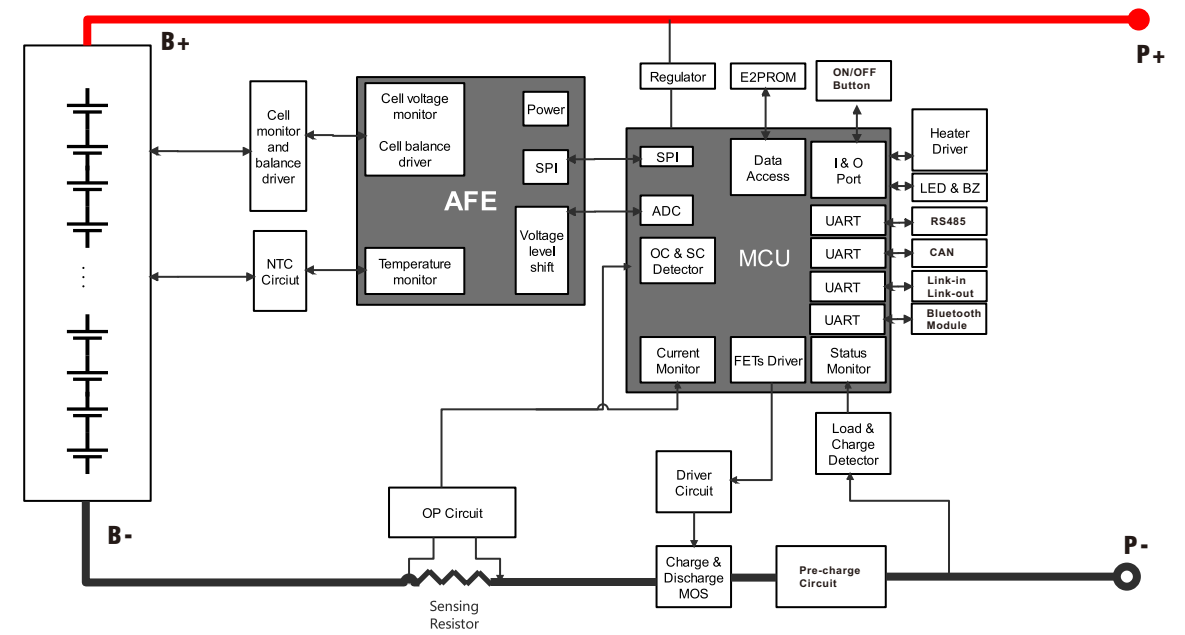
50Ah / 70Ah / 100Ah / 135Ah
150Ah 4U / 150Ah 6U / 200Ah

51.2V Series

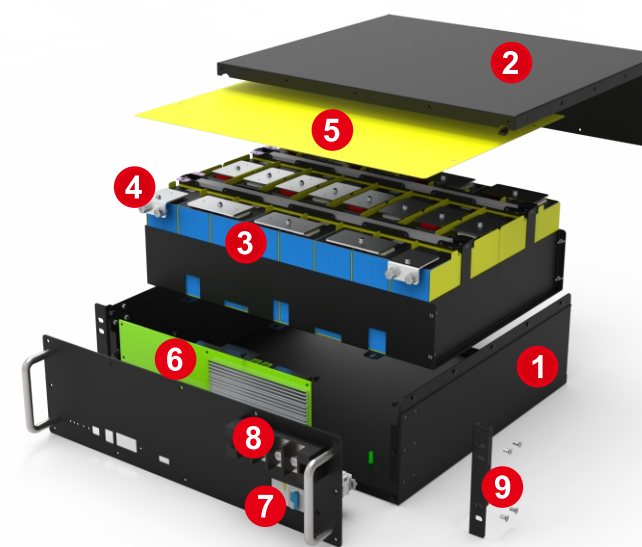
50Ah / 70Ah / 100Ah / 135Ah
150Ah 4U / 150Ah 6U / 200Ah

RESS BATTERY-48V/51.2V

SYSTEM TOPOLOGY



INTERNAL STRUCTURE DIAGRAM



- ① Battery Case: SPCC with black painting
- ② Battery Cover: SPCC with black painting
- ③ LFP Cells (15S for 48V, 16S for 51.2V)
- ④ Aluminum Bar -A1060
- ⑤ Epoxy Board
- ⑥ BMS- Ritar (B0 version)
- ⑦ MCB
- ⑧ Power Terminal
- ⑨ Brackets

RESS BATTERY-48V



PARAMETERS-48V

Model	R-LFP48V50Ah	R-LFP48V70Ah	R-LFP48V100Ah
Nominal Voltage [V]	48	48	48
Nominal Capacity [Ah]	50	70	100
Total Energy [Wh]	2400	3360	4800
Dimension (W*D*H, mm)	442*400*130.5	442*400*130.5	442*400*130.5
Weight [Kg]	26.5	29.6	37.5
Max. Charging Current [A]	50	100	150
Max. Discharging Current [A]	50	100	150
Pulse Discharge Current	55A @10min 60A @10s	77A @10min 84A @10s	110A @10min 120A @10s
Charging Voltage [V]	51.8~54.7		
End of Discharge Voltage [V]	42 (Backup Application) / 45 (Cycle Application)		
Operation Humidity	0~95% RH (No condensing)		
Operating Temperature Range ⁽¹⁾	Charge: 0 ~ +55°C; Discharge: -20 ~ +55°C		
Cycle Life ⁽²⁾	>4000		
Designed Calendar Life	10 Years		
Communication interface	RS485*2, CAN*1,Dry contact *2		
Protection	Over voltage , Low voltage, Over current, Over Temperature, Low Temperature, Short circuit		
Parallel Support ⁽³⁾	Yes, Max. 15 Sets		
Series Support	Not support		
IP Class	IP20		
Certification ⁽⁴⁾	UN38.3, MSDS, CE		

- Note:
- (1) Normally the Li-ion battery operation temperature range is : discharge -20~+55°C, charge 0~+55°C, If the optional heater is stalled, it will automatic start to work once the cell temperature is below 5°C and the heater can help to increase cell temperature 4~8°C/hour.With Optional Heater: Charge / Discharge: -30 ~ +55°C
- (2) 25°C ,0.5C/0.5C, 100%DOD and 80% EOL
- (3) For parallel power cable connection, refer to user manual.
- (4) some models cell have UL , IEC certification,

The datasheet is subject to change without prior notification

RESS BATTERY-48V



PARAMETERS-48V

Model	R-LFP48V135Ah	R-LFP48V150Ah (6U)	R-LFP48V150Ah(4U)	R-LFP48V200Ah
Nominal Voltage [V]	48	48	48	48
Nominal Capacity [Ah]	135	150	150	200
Total Energy [Wh]	6480	7200	7200	9600
Dimension (W*D*H, mm)	442*450*175	442*396*260	442*520*175	442*550*250
Weight [Kg]	58	68	65	78.4
Max. Charging Current [A]	100	100	100	100
Max. Discharging Current [A]	100	100	100	100
Pulse Discharge Current	110A @10min 120A @10s	110A @10min 120A @10s	110A @10min 120A @10s	110A @10min 120A @10s
Charging Voltage [V]	51.8~54.7			
End of Discharge Voltage [V]	42 (Backup Application) / 45 (Cycle Application)			
Operation Humidity	0~95% RH (No condensing)			
Operating Temperature Range ⁽¹⁾	Charge: 0 ~ +55°C; Discharge: -20 ~ +55°C			
Cycle Life ⁽²⁾	>4000			
Designed Calendar Life	10 Years			
Communication interface	RS485*2, CAN*1, Dry contact *2			
Protection	Over voltage , Low voltage, Over current, Over Temperature, Low Temperature, Short circuit			
Parallel Support ⁽³⁾	Yes, Max. 15 Sets			
Series Support	Not support			
IP Class	IP20			
Certification ⁽⁴⁾	UN38.3, MSDS, CE			

- Note:
- (1) Normally the Li-ion battery operation temperature range is : discharge -20~+55°C, charge 0~+55°C, If the optional heater is stalled, it will automatic start to work once the cell temperature is below 5°C and the heater can help to increase cell temperature 4~8°C/hour.With Optional Heater: Charge / Discharge: -30 ~ +55°C
- (2) 25°C ,0.5C/0.5C, 100%DOD and 80% EOL
- (3) For parallel power cable connection, refer to user manual.
- (4) some models cell have UL , IEC certification,

RESS BATTERY-51.2V



PARAMETERS-51.2V

Model	R-LFP51.2V50Ah	R-LFP51.2V70Ah	R-LFP51.2V100Ah
Nominal Voltage [V]	51.2	51.2	51.2
Nominal Capacity [Ah]	50	70	100
Total Energy [Wh]	2560	3584	5120
Dimension (W*D*H, mm)	442*400*130.5	442*400*130.5	442*400*130.5
Weight [Kg]	27.8	31.0	39.5
Max. Charging Current [A]	50	100	150
Max. Discharging Current [A]	50	100	150
Pulse Discharge Current	55A @10min 60A @10s	77A @10min 84A @10s	110A @10min 120A @10s
Charging Voltage [V]	55.2~58.4		
End of Discharge Voltage [V]	44.8 (Backup Application) / 48 (Cycle Application)		
Operation Humidity	0~95% RH (No condensing)		
Operating Temperature Range ⁽¹⁾	Charge: 0 ~ +55°C; Discharge: -20 ~ +55°C		
Cycle Life ⁽²⁾	>4000		
Designed Calendar Life	10 Years		
Communication interface	RS485*2, CAN*1,Dry contact *2		
Protection	Over voltage , Low voltage, Over current, Over Temperature, Low Temperature, Short circuit		
Parallel Support ⁽³⁾	Yes, Max. 15 Sets		
Series Support	Not support		
IP Class	IP20		
Certification ⁽⁴⁾	UN38.3, MSDS, CE		

- Note:
- (1) Normally the Li-ion battery operation temperature range is : discharge -20~+55°C, charge 0~+55°C, If the optional heater is stalled, it will automatic start to work once the cell temperature is below 5°C and the heater can help to increase cell temperature 4~8°C/hour.With Optional Heater: Charge / Discharge: -30 ~ +55°C
- (2) 25°C ,0.5C/0.5C, 100%DOD and 80% EOL
- (3) For parallel power cable connection, refer to user manual.
- (4) some models cell have UL , IEC certification,

RESS BATTERY-51.2V



PARAMETERS-51.2V

Model	R-LFP51.2V135Ah	R-LFP51.2V150Ah (6U)	R-LFP51.2V150Ah(4U)	R-LFP51.2V200Ah
Nominal Voltage [V]	51.2	51.2	51.2	51.2
Nominal Capacity [Ah]	135	150	150	200
Total Energy [Wh]	6912	7680	7680	10240
Dimension (W*D*H, mm)	442*450*175	442*396*260	442*520*175	442*550*250
Weight [Kg]	60.7	71.6	68	82.4
Max. Charging Current [A]	100	100	100	100
Max. Discharging Current [A]	100	100	100	100
Pulse Discharge Current	110A @10min 120A @10s	110A @10min 120A @10s	110A @10min 120A @10s	110A @10min 120A @10s
Charging Voltage [V]	55.2~58.4			
End of Discharge Voltage [V]	44.8 (Backup Application) / 48 (Cycle Application)			
Operation Humidity	0~95% RH (No condensing)			
Operating Temperature Range ⁽¹⁾	Charge: 0 ~ +55°C; Discharge: -20 ~ +55°C			
Cycle Life ⁽²⁾	>4000			
Designed Calendar Life	10 Years			
Communication interface	RS485*2, CAN*1, Dry contact *2			
Protection	Over voltage , Low voltage, Over current, Over Temperature, Low Temperature, Short circuit			
Parallel Support ⁽³⁾	Yes, Max. 15 Sets			
Series Support	Not support			
IP Class	IP20			
Certification ⁽⁴⁾	UN38.3, MSDS, CE			

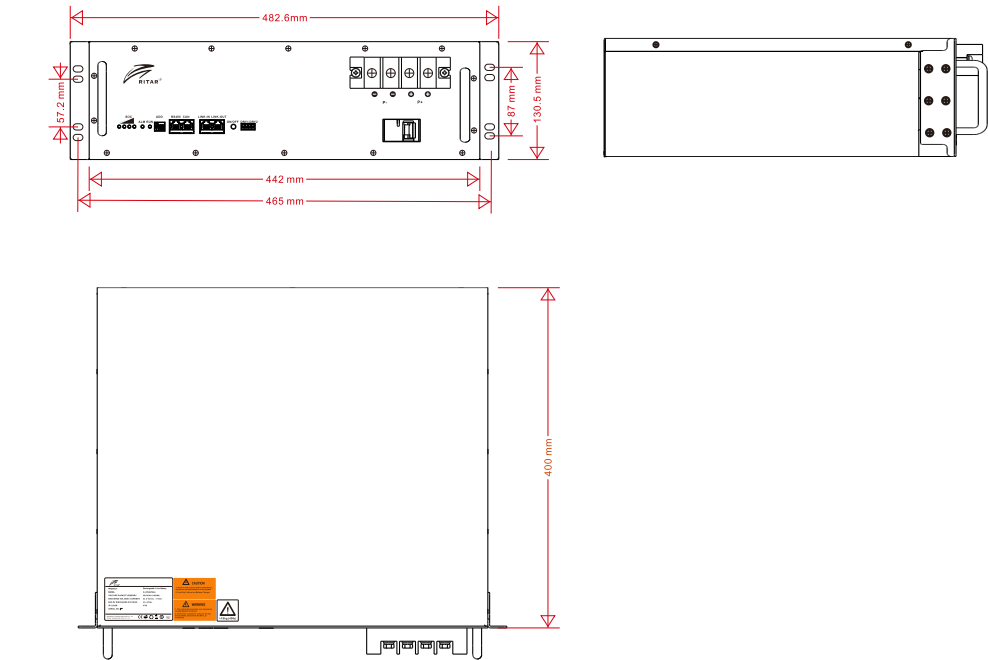
- Note:
- (1) Normally the Li-ion battery operation temperature range is : discharge -20~+55°C, charge 0~+55°C, If the optional heater is stalled, it will automatic start to work once the cell temperature is below 5°C and the heater can help to increase cell temperature 4~8°C/hour.With Optional Heater: Charge / Discharge: -30 ~ +55°C
- (2) 25°C ,0.5C/0.5C, 100%DOD and 80% EOL
- (3) For parallel power cable connection, refer to user manual.
- (4) some models cell have UL , IEC certification,

RESS BATTERY-48V

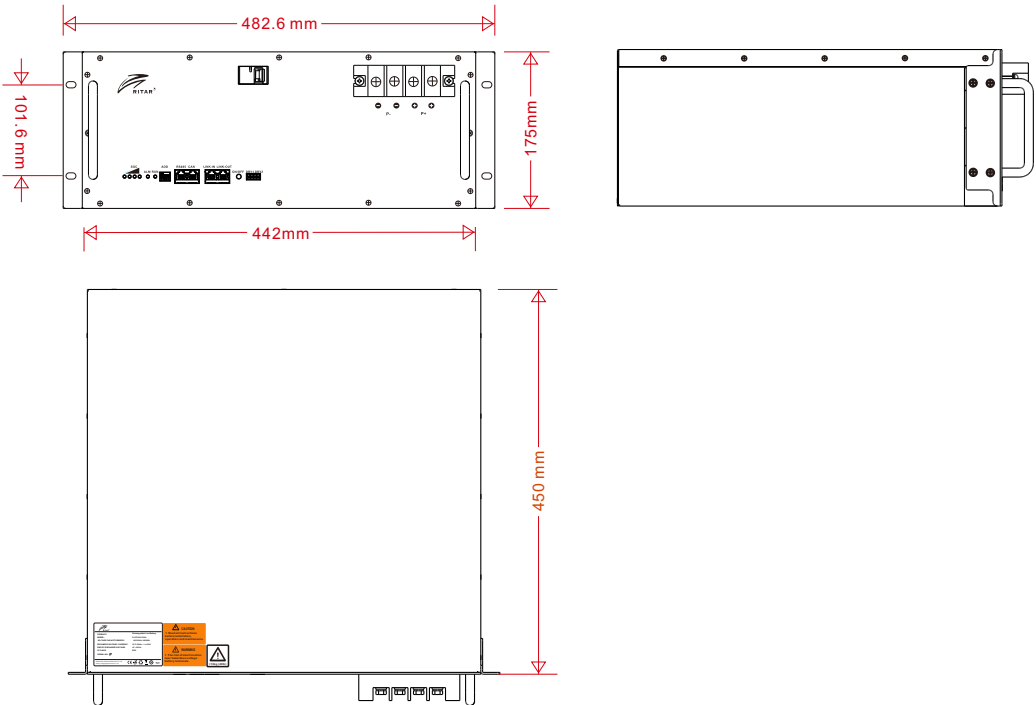


Dimension

R-LFP48V50Ah (Without MCB) /R-LFP51.2V50Ah (Without MCB)
R-LFP48V70Ah /R-LFP51.2V70Ah
R-LFP48V100Ah /R-LFP51.2V100Ah



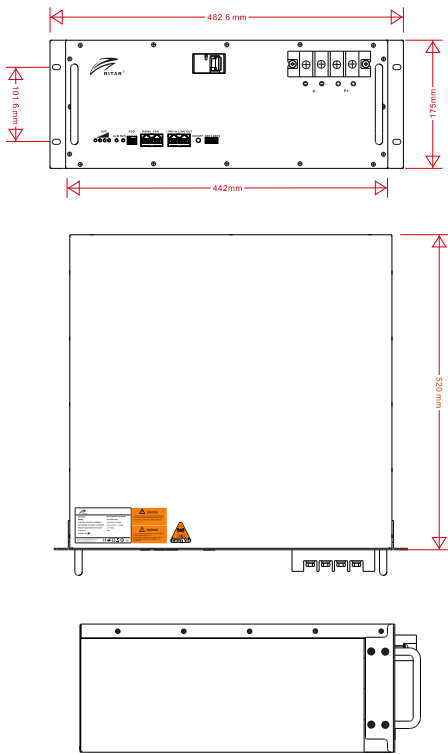
R-LFP48V135Ah /R-LFP51.2V135Ah



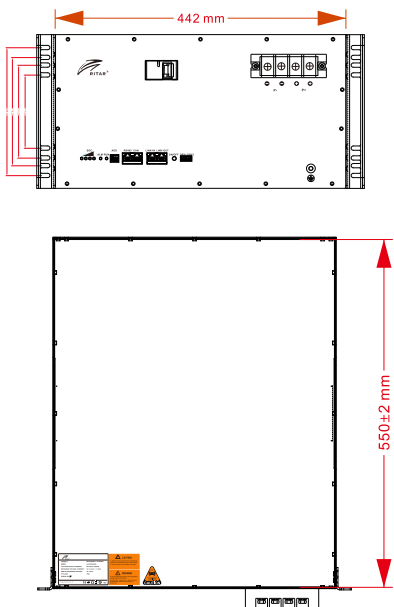
RESS BATTERY-48V



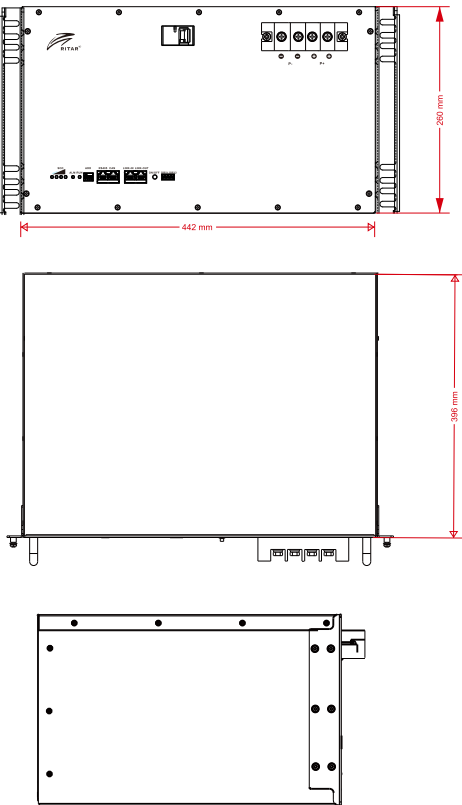
R-LFP48V150Ah 4U
R-LFP51.2V150Ah 4U



R-LFP48V200Ah / R-LFP51.2V200Ah



R-LFP48V150Ah 6U
R-LFP51.2V150Ah 6U

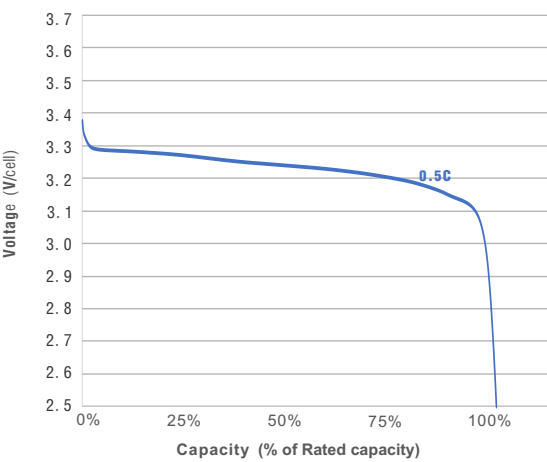


RESS BATTERY-48V/51.2V

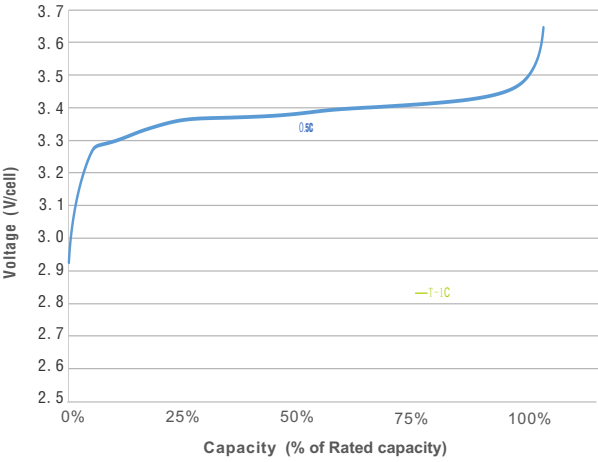


Performance Curve

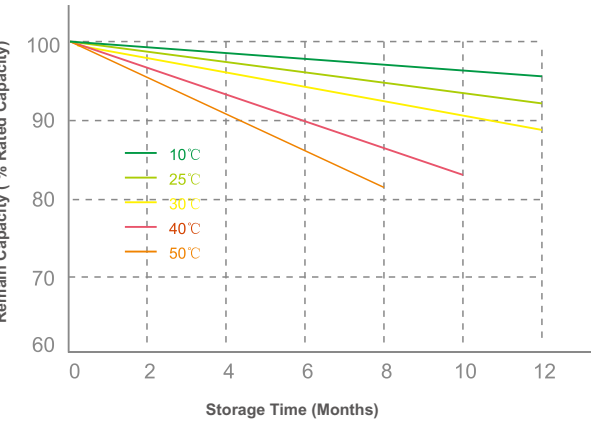
Dierent Discharge Rate and Temperature Characteristic



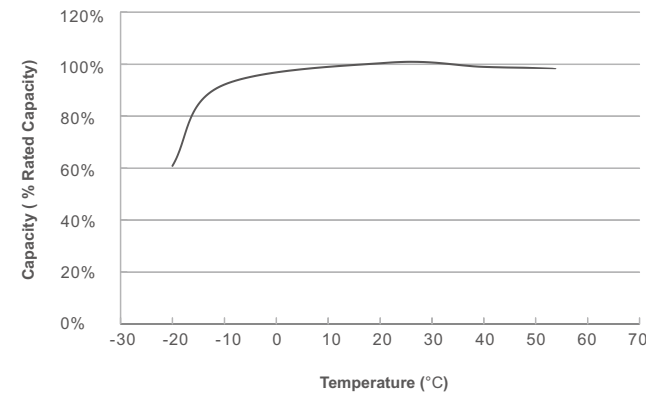
Dierent Charge Rate and Temperature Characteristic



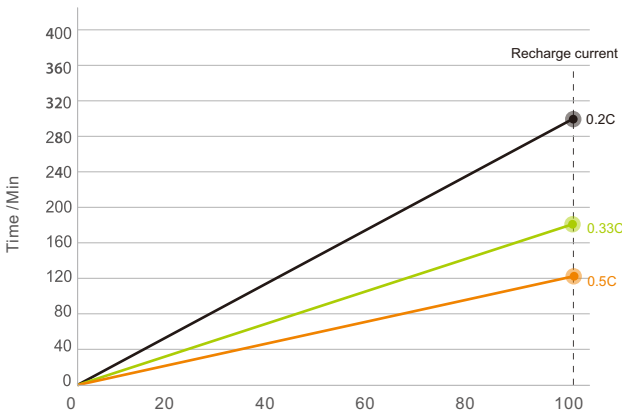
Different Temperature Self Discharge Curve



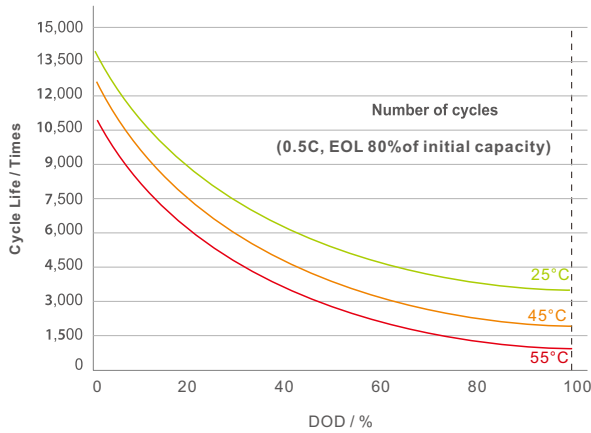
Capacity with Different Temperature



Typical Recharge Time



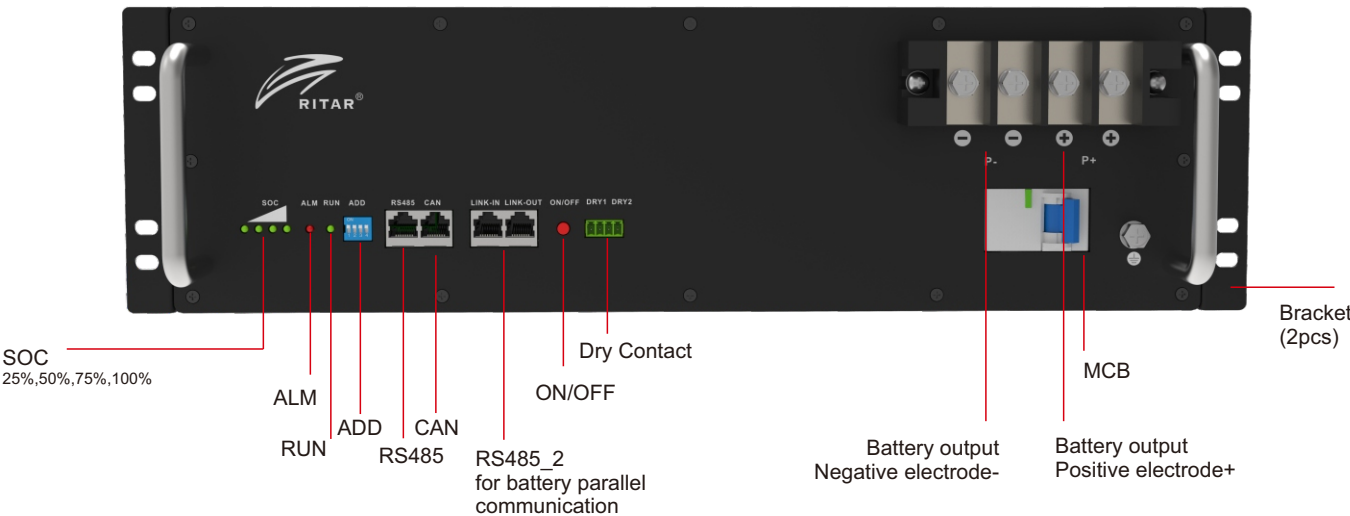
Typical Cycle Life



RESS BATTERY-48V/51.2V



Front Panel



Optional Parts: Bluetooth module & APP

